

SAINT JOHN, NB, Canada

WMO: 716090

Lat: 45.318N

Lon: 65.886W

Elev: 109

StdP: 100.02

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-19.3	-28.5	0.3	-20.8	-25.8	0.4	-18.6	15.5	3.4	13.4	-1.2	3.4	0	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	26.3	18.9	24.7	18.0	23.2	17.2	20.5	24.1	19.4	22.7	18.5	21.5	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	21.9	18.2	13.3	20.8	17.4	12.6	19.8	59.5	24.3	55.9	22.8	52.8	21.4	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	10.9	9.6	-25.6	30.1	2.5	1.4	-27.4	31.1	-28.9	31.9	-30.3	32.7	-32.1	33.7		
			-25.9	22.5	2.5	1.0	-27.7	23.2	-29.1	23.7	-30.5	24.3	-32.3	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-7.5	-6.5	-2.2	3.9	9.6	14.2	17.5	17.3	13.6	8.1	2.6	-3.2	
	DBStd	9.67	6.54	5.92	4.98	3.38	2.96	2.91	2.14	2.23	3.41	3.80	4.79	5.63	
	HDD10.0	2338	542	462	377	184	44	3	0	0	8	84	225	408	
	HDD18.3	4651	800	696	636	433	272	126	42	46	144	318	471	667	
	CDD10.0	764	0	0	0	1	31	131	232	226	117	24	4	0	
	CDD18.3	36	0	0	0	0	0	3	16	13	3	0	0	0	
	CDH23.3	314	0	0	0	0	6	48	134	104	21	1	0	0	
	CDH26.7	41	0	0	0	0	1	6	18	12	4	0	0	0	
Wind	WSAvg	4.3	4.8	4.9	5.0	4.9	4.3	3.9	3.4	3.3	3.7	4.3	4.6	4.8	
Precipitation	PrecAvg	1262	110	99	104	91	99	96	88	76	111	129	130	130	
	PrecMax	1771	180	215	207	171	226	227	181	167	242	240	222	270	
	PrecMin	764	37	31	17	27	37	21	25	11	31	27	43	33	
	PrecStd	218	41	42	42	38	44	55	46	43	61	57	48	54	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	8.9	12.0	16.8	23.6	27.6	28.9	28.3	25.8	20.7	16.2	11.8	
		MCWB	9.7	7.6	6.7	10.2	15.0	18.8	20.5	20.1	18.8	15.5	14.5	11.0	
	2%	DB	7.3	5.8	8.6	13.9	20.3	24.6	26.3	25.8	22.9	17.6	14.0	9.6	
		MCWB	6.7	4.6	5.6	8.9	13.2	17.0	19.0	18.9	17.6	14.7	12.9	8.9	
	5%	DB	4.8	3.8	6.3	11.8	17.9	22.4	24.8	24.3	21.0	15.8	12.1	7.5	
		MCWB	4.0	2.6	4.0	7.6	11.6	15.7	18.2	17.9	16.5	13.4	10.8	6.4	
	10%	DB	2.4	2.3	4.4	9.9	15.8	20.5	23.1	22.8	19.3	14.3	10.3	5.1	
		MCWB	1.4	1.1	2.5	6.3	10.9	14.7	17.4	17.3	15.8	12.2	9.0	3.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	8.0	9.8	12.7	16.9	20.0	22.0	21.9	20.7	17.6	15.2	11.2
			MCDB	10.4	8.7	11.0	14.0	21.3	25.3	26.9	26.1	23.8	19.1	15.9	11.5
		2%	WB	6.8	4.8	6.4	10.5	14.8	18.4	20.5	20.4	19.1	15.7	13.2	9.1
			MCDB	7.2	5.5	7.6	12.4	17.9	22.6	24.5	23.7	21.1	16.9	13.8	9.6
5%		WB	4.1	2.8	4.4	8.6	13.1	17.1	19.4	19.4	17.8	14.2	11.1	6.8	
		MCDB	4.6	3.6	5.8	10.7	16.3	20.5	22.9	22.3	19.7	15.2	11.8	7.4	
10%		WB	1.4	1.2	2.7	6.9	11.7	15.9	18.4	18.3	16.7	12.6	9.1	4.2	
		MCDB	2.3	2.2	4.2	9.3	14.6	18.9	21.6	21.1	18.5	13.8	10.1	5.1	
Mean Daily Temperature Range		5% DB	MDBR	9.9	9.9	8.9	9.1	10.2	10.4	10.1	10.4	10.1	9.0	8.0	8.4
			MCDBR	12.4	9.2	10.7	12.8	14.7	14.1	13.7	13.5	12.0	10.9	9.4	9.8
	MCWBR		12.9	8.7	8.4	8.5	8.7	7.9	7.4	7.4	7.4	8.1	9.0	10.1	
	5% WB	MCDBR	12.3	9.2	9.3	10.3	11.5	11.2	10.8	10.3	9.1	8.7	8.5	9.7	
		MCWBR	12.8	9.2	8.4	8.2	7.9	7.2	6.7	6.5	6.5	7.5	8.9	10.3	
Clear-Sky Solar Irradiance	taub	0.284	0.290	0.319	0.350	0.373	0.410	0.431	0.405	0.374	0.331	0.309	0.295		
	taud	2.452	2.442	2.436	2.398	2.381	2.309	2.276	2.356	2.410	2.527	2.533	2.459		
	Ebn at Noon	837	901	916	910	895	859	835	846	844	842	802	781		
	Edh at Noon	69	84	98	111	117	126	129	115	100	78	65	63		
All-Sky Solar Radiation	RadAvg	1.45	2.31	3.47	4.33	4.99	5.42	5.54	4.96	3.99	2.53	1.45	1.10		
	RadStd	0.13	0.18	0.23	0.40	0.56	0.46	0.40	0.33	0.29	0.26	0.16	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	+0.74	+1.03	N/A	N/A	+42	N/A	
Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.34	+0.46	N/A	N/A	+54	N/A	

Nomenclature: See separate page